Gokturk Colak

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Education

Istanbul University — **Beyazit, Istanbul**
Bachelor's in Economics, English Program — ***August 2022 - Present***

- **GPA: 3.70/4.00**
- **Relevant Coursework:** Econometrics, Statistics, Mathematical Economics, Data Structures (Python), Data Analysis (R), Machine Learning, Artificial Intelligence, Blockchain Principles

University of Tübingen — **Tübingen, Germany**
Erasmus+ Exchange Program, Economics — ***April 2026 - September 2026***

- **Relevant Coursework:** Quantitative Analysis (R), Time Series Analysis, Monte Carlo Inference, Statistical Inference, MLE, OLS/Logit, Data Science (Python and R), Topics in Crypto

Skills

Programming Languages - Python, R
Tools - Excel, RStudio, VS Code, Jupyter, Git, GitHub
Machine Learning - Scikit-Learn, XGBoost, LightGBM, TensorFlow, Keras, caret, tidymodels
Languages - Turkish (Native), English (Advanced), German (Beginner)
Certifications - Machine Learning for Time Series Data (DataCamp), Exploratory Data Analysis (DataCamp), Next Generation Banking Simulation (Isbank), Digital Banking (Isbank)

Experience

Istanbul University Informatics Department — **Vezneciler, Istanbul**
Undergraduate Research Assistant — ***November 2023 - June 2024***

- Worked in a 3-person team to build a complete database of MIS department heads across all 81 provinces
- Collected, cleaned, and standardized research datasets in Excel and Google Sheets

Projects

Localify — **Slush'D Hackathon**
Generative AI, Computer Vision, LLMs, Prompt Engineering — ***October 2026***

- Co-built a vision-model pipeline (fal, Apify) that flags which Instagram posts need localization for each market
- Developed prompts for an LLM agent (Manus API) returning per-market change plans as structured JSON; open-sourced

Bitcoin's 21M Cap vs. Inflation Targeting — **Istanbul University**
Seminar Paper — ***April - August 2026***

- Assessed whether Bitcoin's fixed supply solves the time-inconsistency problem addressed by inflation targeting (Kydland–Prescott, Barro–Gordon, Bernanke–Mishkin), using interwar gold standard evidence
- Concluded that a fixed supply strengthens commitment but is an inferior monetary rule due to debt-deflation risk, price volatility, and a shrinking fee base for network security

Credit Risk and Econometric Modeling in R — **Tübingen University**
R, Logit, Monte Carlo Simulation — ***August 2026***

- Compared LPM vs. Logit (average partial effects, robust SEs) on 45,000 loan applications, diagnosing quasi-complete separation
- Validated estimator bias/variance/MSE and test power curves via Monte Carlo simulation against asymptotic theory; implemented Bernoulli MLE via numerical optimization
- Modeled China's CO2 emissions growth with AR models using ADF tests, AIC/BIC lag selection, and Ljung-Box diagnostics

Algorithmic Trading System
Python, XGBoost, Binance API, AWS EC2 — ***August - December 2025***

- Trained an XGBoost classifier on RSI, MACD, and OBV indicators to filter trade signals, validated through backtesting
- Implemented hysteresis-based entry/exit logic to hold momentum positions and filter volatility; deployed on AWS EC2

Activities

Istanbul University Young Investors Club
Researcher — ***October 2025 - Present***

- Conduct research and draft analyses supporting the club's weekly social media insights

Istanbul University Data Club
Vice President — ***October 2023 - October 2024***

- Organized an institutional visit to IBB Data Laboratory and managed corporate communications